Fiscal Year:	FY 2020	**Task Last Updated:**	FY 07/09/2020
PI Name:	Turek, Fred W Ph.D.		
Project Title:	Effects of Spaceflight on Gastrointestinal Microbiota in Mice: Mechanisms and Impact on Multi-System Physiology		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Evanston	**State:**	IL
Zip Code:	60201-2919	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	06/01/2015	**End Date:**	05/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 5/31/2021 per F. Hernandez/ARC and NSSC information (Ed., 7/17/20) NOTE: End date changed to 5/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 6/14/19) NOTE: End date changed to 5/31/2019 per NSSC information (Ed., 6/7/18)		
Key Personnel Changes/Previous PI:	As of 2017, Dr. Peng Jiang is a Research Assistant Professor at Northwestern (formerly a post-doc); his involvement in the project continues. Dr. Robin Voigt-Zuwalla at Rush University Medical Center is joining the Keshavarzian team examining GI function. Additional investigators at Northwestern will join the project: Dr. Tiffany Schmidt will evaluate eye/retina physiology; Dr. Lisa Wilsbacher will evaluate heart/cardiovascular physiology; Dr. Teresa Woodruff will evaluate reproductive tract physiology; Dr. Joseph Bass will evaluate pancreas and muscle metabolic physiology; Dr. Phyllis Zee will evaluate brain gene expression. Dr. Keith Summa MD PhD will be joining the team to evaluate GI tissue samples.		

COI Name (Institution):	Green, Stefan Ph.D. (University of Illinois) Keshavarzian, Ali M.D. (Rush University Medical Center) Forsyth, Christopher Ph.D. (Rush University Medical Center) Vitaterna, Martha Ph.D. (Northwestern University) Jiang, Peng Ph.D. (Northwestern University)
Grant/Contract No.:	NNX15AL05G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This proposal in response to NNH14ZTT002N NRA “Research Opportunities for Flight Experiments in Space Biology” is intended to examine the potential role(s) of disruption of microbial communities in the gut (dysbiosis) in mammalian adjustment to the space environment. Many challenges to mammalian physiological homeostasis present in the space environment are known to create dysbiosis. Similarly, immune or inflammatory changes, as well as metabolic alterations, observed with time in spaceflight may be attributable in part to dysbiosis. A series of experiments in mice will be carried out to define the mechanisms by which time in space, diet, and host genotype interact to impact the composition of the gut microbiota, and how dysbiosis relates to gene expression (by RNA-seq) and physiology in serum, colon, ileum, spleen, liver, and fat as well as the sleep/wake cycle and feeding behavior. Extensive use of ground based studies will support the analysis and interpretation of spaceflight data. Five projects at three institutions will focus on different aspects. Project 1 (University of Illinois at Chicago, Principal Investigator (PI: SJ Green) will carry out metagenomic sequencing of microbiota from fecal samples and cecal contents (including from the tissue sharing opportunity); Projects 2 and 3 (Rush University Medical Center, PIs: A Keshavarzian and C Forsyth) will use gene expression, physiological and histologic measures of gastrointestinal and immune function, respectively; while Projects 4 and 5 (Northwestern University, PIs: FW Turek and MH Vitaterna) will focus on metabolism, sleep, and circadian behavioral organization. Dr. Turek (Northwestern) will provide Principal Investigator (PI) oversight. These studies are in alignment with NASA Research Announcement Research Emphases 1)“Spaceflight Omics Studies” and 3)“Understanding How Complex Organisms Adapt to Spaceflight.”
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal will examine the impact of the space environment on the population structure of the intestinal microbiota of mice, and how multiple physiological systems involving the gastrointestinal (GI), immune, metabolic, circadian, and sleep systems, known to be affected by the microbiota, are impacted by the space environment. While our overall working hypothesis is that the space environment and associated stressors will have a negative impact on the microbiota (i.e., dysbiosis), and that this in turn will negatively impact multiple physiological systems known to be adversely affected by spaceflight, our integrated analysis of gene expression and multiple tissue/organ specific biomarkers of tissue/organ injury will also allow us to uncover the adverse effects of spaceflight due or not due to dysbiosis. However, the impact of the proposed studies extends beyond understanding the mechanisms at play in the unique stresses of spaceflight. The proposed studies will elucidate mechanisms underlying interactions between GI, immune, metabolic, and sleep functions and specifically the role of the microbiota in these interactions. This knowledge has tremendous potential for guiding development of dysbiosis-targeted interventions for disorders in all of these systems.
Task Progress:	The ultimate goal of this study is to determine the effects of spaceflight over time on an interconnected set of physiological systems: 1) the gut microbiota; 2) gastrointestinal function; 3) immune function; 4) metabolism; and 5) sleep and circadian behavioral organization. Rodent Research 7 (RR7) compares mice of two different genotypes (inbred strains): C57BL/6J (the most widely-used mouse strain) and C3H/HeJ. The study also included comparisons of two different durations of spaceflight (~25 and ~75 days). In addition to the Flight group, three ground control groups were also part of the study: Basal (representing the pre-launch state), Vivarium (standard vivarium housing for the same duration of time as flight), and Ground (same habitat in the International Space Station Environment Simulator, ISSES). Twenty mice (10 of each strain) were included in each experimental group. All mice completed the study without significant health issues. Samples were obtained from a total of 80 mice. In addition, fecal samples, food samples, and environmental swabs were obtained every 2 weeks from the Vivarium, Ground, and Flight conditions. Three 48-hour continuous video recording sessions were obtained from the six cages/habitats of these conditions as well, at approximately monthly intervals. Advances in the past year: 1) We developed, validated, and applied a new statistical method for comparing microbiome changes, which revealed reproducible spaceflight-induced microbiome changes from RR-1 and STS-135 missions. These findings have been published in Microbiome (Jiang et al., 2019). 2) In collaboration with GeneLab, we validated our specimen processing methods for RR-7 tissues that were preserved in RNA-later. 3) We completed a first-pass analysis of the behaviors of mice inside the ground and the flight habitats. 4) We validated our specimen processing methods for RR-7 microbiome samples and DNA extractions from 381 samples are complete. Issues encountered in the past year: 1) We learned that the light spectrum of the flight and the ground habitats did not match ; 2) A re-run of the ground control animals including both blue-enriched and white light habitats is planned ; 3) To avoid batch effects, RNA-seq analysis will not be done until samples from these additional control animals can be included ; 4) Metagenomic sequencing for microbiome analysis was delayed by laboratory closures due to COVID-19.
Bibliography Type:	Description: (Last Updated: 07/16/2025)

Articles in Other Journals or Periodicals	Vitaterna MH, Jiang P. "The microbiome in space, from the Apollo missions to present." Physiology News 2020 Jan;117:16-7. https://doi.org/10.36866/pn.117.16 , Jan-2020
Articles in Peer-reviewed Journals	Jiang P, Green SJ, Chlipala GE, Turek FW, Vitaterna MH. "Reproducible changes in the gut microbiome suggest a shift in microbial and host metabolism during spaceflight." Microbiome. 2019 Aug 9;7(1):113. https://doi.org/10.1186/s40168-019-0724-4 ; PMID: 31399081; PMCID: PMC6689164 , Aug-2019
Significant Media Coverage	Daley J. "Spaceflight Alters the Gut Microbes of Mice and Men. Article on the PI's research and the Microbiome Project." Scientific American. September 5, 2019., Sep-2019